To:
Editor-in-Chief
Biomedical Signal Processing and Control (BSPC)
Elsevier

Subject: Submission of Original Research Manuscript

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research manuscript titled:
"End-to-End Deep Representation Learning and Computer Vision for Objective Hearing Screening via Auditory-Evoked Pupillary Responses"
for consideration as a Full Length Article in Biomedical Signal Processing and Control (BSPC).

Universal newborn and pediatric hearing screening currently relies on Automated Auditory Brainstem Responses (AABR) and Otoacoustic Emissions (TEOAE), both of which suffer from practical bottlenecks: skin abrasion, probe dislodgement, susceptibility to ear canal vernix, and expensive hardware. The Auditory-Evoked Pupillary Response (AEPR), governed by the subcortical Locus Coeruleus-Norepinephrine (LC-NE) system, provides a non-invasive optical alternative. However, its adoption has been hampered by heuristic thresholds, vulnerability to motion artifacts, and subject leakage in prior studies.

In this work, we present an end-to-end, reproducible deep learning and computer vision framework for single-trial acoustic salience detection:
1. Standardized, Leakage-Free Benchmark: Across three multi-site cohorts (N=66, N=19, N=6; 21,164 single-trial epochs), we enforce strictly subject-independent 5-fold cross-validation (StratifiedGroupKFold).
2. Deep Signal Architecture SOTA: Benchmarking classical models against Multi-Scale 1D-CNNs, Dilated TCNs, Bi-LSTMs with Bahdanau attention, and CNN-Transformers, our CNN-Transformer achieves state-of-the-art ROC-AUC of 0.844 (95% CI: [0.835, 0.853]) and PR-AUC of 0.512 (a 4.2x gain over the 0.122 chance baseline; DeLong paired test Z = 9.48, p < 10^-15).
3. Physiological Saliency Discovery: Temporal self-attention autonomously isolates the primary discriminative window at t in [0.8s, 2.2s], faithfully mirroring human LC-NE autonomic dilation kinetics.
4. 50% Latency Reduction: Observation windows can be truncated to t = 1.5s while preserving ~96% of maximal screening accuracy.
5. Commodity Hardware Feasibility: Direct least-squares ellipse fitting on 30 fps eye video achieves Pearson r = 0.991 (p < 10^-15) against research-grade infrared eye-trackers.

This manuscript aligns squarely with the scope of Biomedical Signal Processing and Control, specifically advancing physiological time-series modeling, signal decontamination, and neural network classification for diagnostic biosignals.

This work is entirely original, has not been published elsewhere, and is not currently under consideration by any other journal. All authors have approved the submission.

Thank you for your time and consideration of our work.

Sincerely,

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